

# LARP: Local Anticipatory Routing Protocol — Preregistration: Human Judgment of Local AI Model Solvability

ANONYMOUS AUTHOR(S)

## 1 Motivation

LLM use is now dominated by routine, low-complexity requests — writing assistance, information-seeking, practical guidance — rather than the symbolic or mathematical problems that most benefit from a frontier model [1]. Small models running locally on consumer hardware already reach satisfactory benchmark accuracy on exactly this class of routine task, at a fraction of the computational and environmental cost [2, 3, 4]. Yet consumer interfaces provide no mechanism for exploiting this: every request is sent to the cloud by default, and users receive no signal about whether a given prompt could have been handled locally [5].

One route toward closing this gap is to let users make the routing decision themselves — an interface where people choose, per request, whether to stay local or go to the cloud. This is only viable if people can actually tell the difference. If users cannot judge which prompts a small local model can handle, self-routing is not a workable interaction, and the decision must instead be made for them automatically. This planned study addresses that precondition directly:

**RQ:** How well can people judge whether a prompt can be answered acceptably by a small, locally-run language model, compared to a ground-truth solvability label?

## 2 Study Design

### 2.1 Hypotheses

- H1: Users’ judgments of local solvability diverge from the model’s actual performance; agreement with ground truth is at best “slight” (following Landis and Koch [6],  $\kappa_w \leq 0.20$ ). This is tested against a preregistered adequacy floor of  $\kappa_{\min} = 0.50$  (see *Analysis*).
- H2 (exploratory): The disagreement is directional, with users routing more prompts to the cloud.
- H3 (exploratory): The disagreement is directional, with users routing more prompts to the local model.

### 2.2 Independent Variables

Single Group Design.

- Prompts are presented to participants via stratified random sampling.
- The sample is not forced onto a fixed local solvable balance.
- Stratifying on  $\hat{p}$  ensures as close to the full difficulty range of the dataset (see *Ground Truth*) as possible is covered rather than reproducing its natural composition. The sample is therefore deliberately *not* proportional to the pool, and participants are not told the class balance to avoid inferring the solvability distribution of their question pool.

### 2.3 Dependent Variables

- Human six-point rating per prompt and participant.

## 2.4 Descriptive & Exploratory Variables

- Gender
- Age
- Educational Background
- Occupation
- Task Time
- GenAI usage frequency (self-reported)

## 2.5 Questionnaires

*Pre-Study (Screening).*

- Prior LLM exposure: included if the participant has used a general-purpose LLM chat interface at least once. We deliberately do *not* screen on usage frequency (see below);
- frequency is recorded post-study as an exploratory covariate.

*Response scale.* We use a six-point, midpoint-free Likert scale whose levels partition the probability interval  $[0, 1]$  into equal sixths:

$$\underbrace{\text{Extremely unlikely, Very unlikely, Unlikely}}_{\text{cloud (0,1,2)}} \mid \underbrace{\text{Likely, Very likely, Extremely likely}}_{\text{local (3,4,5)}}$$

Each level is presented with its explicit percentage band — *Extremely unlikely* (0–17 %), *Very unlikely* (17–33 %), *Unlikely* (33–50 %), *Likely* (50–67 %), *Very likely* (67–83 %), *Extremely likely* (83–100 %) — and the scale is explained in the task instructions before any rating is made. Because the bands are of equal width, their midpoints are equally spaced, which is what licenses the linearly weighted  $\kappa_w$  used in the analysis. Framing the scale in probability bands rather than “ $k$  in 5” counts also decouples the human instrument from the number of generations used to establish ground truth, so the scale does not change if  $k$  does. The three-versus-three split places the routing boundary at  $p = 0.5$ , i.e. at “more likely than not”. The lack of a midpoint avoids the central-tendency default that plagues scales with a neutral zero, and—since this is a performance study, not a study of self-routing behaviour—an abstention would add a variable without adding signal. We elicit the instinctive judgement directly:

“If the local model were asked to solve this prompt, how likely is it that the local model Gemma4-e2b would produce a satisfactory answer?”

*Why frequency is a covariate, not a filter.* Screening for heavy use assumes frequent users judge local capability more accurately. That assumption is untested and may run backwards: heavy users of frontier cloud models may carry a stronger prior that adequate answers require a large model, and therefore *under*-rate the local model, whereas infrequent users lack that anchor. Since we record frequency regardless, we treat it as an exploratory predictor of both judgment accuracy and its *direction* (the H2/H3 cloud-vs-local bias), rather than as an eligibility gate.

*Post Study.* The demographic and usage items listed under *Descriptive & Exploratory Variables* (gender, age, educational background, occupation, and GenAI usage frequency) are collected here.

## 2.6 Analysis

All analyses are conducted in Python. The unit of analysis is the prompt ( $n = 280$ ), not the individual rater or their judgment. Ratings from the raters of a given prompt are aggregated into a single per-prompt value, and all tests run on these per-prompt values.

Our confirmatory claim is that human raters are unreliable routers, and hence that an automated router (classifier) is warranted in their place. This is tested by H1, which asks whether raters' judgments agree with the ground-truth routing label at at best "slight" rates. H1 is non-directional (it concerns the magnitude of disagreement, not its sign) and is tested at  $\alpha = 0.05$ .

We additionally report two exploratory, directional hypotheses that characterize *how* raters err, conditional on their erring. H2 states that raters systematically route to the cloud more often than needed (underestimating local models); H3 states that they route too locally (overestimating them). These are opposite tails of the same signed quantity, so at most one can be rejected; the only joint outcome is that neither is rejected, indicating no detectable systematic direction. Because we are willing to interpret a rejection in either direction, we test H2 and H3 at  $\alpha = 0.05/2 = 0.025$  each (Bonferroni), which is equivalent to a single two-sided test at  $\alpha = 0.05$ .

We emphasize that H2/H3 are exploratory and do not gate the confirmatory claim: whether or not a systematic direction is detected, H1 alone determines whether raters are unreliable. The direction, if any, informs the design of downstream guidance but is not required to motivate the classifier.

*Related to H1.* Because the six-point human scale and the ground truth  $\hat{p}$  live on the *same* six-level ordinal scale, we tackle H1 through *agreement*.

- **Primary metric:** linearly weighted Cohen's  $\kappa_w$  between the per-prompt aggregated human rating and the ground-truth level, both on the same six-level equal-width scale. The weights are linear because the six bands are of equal width and therefore equally spaced.
- **Threshold test.** One-sided at  $\alpha = 0.05$  (a single confirmatory test, no multiplicity correction):  $H_0: \kappa_w \geq \kappa_{\min}$  vs.  $H_1: \kappa_w < \kappa_{\min}$ , with  $\kappa_{\min} = 0.50$ . This floor is set below the maximum  $\kappa_w$  attainable by a perfectly calibrated rater under our labelling scheme ( $\approx 0.87$  at  $k = 20$ ; see *Ground Truth*), so that a rejection cannot be attributed to measurement noise in the ground-truth label. We report observed  $\kappa_w$  both raw and as a fraction of that bound. Failing to reject  $H_0$  is *inconclusive* and must not be read as evidence for  $\kappa_w \geq \kappa_{\min}$ .
- **Why not correlation.** Correlation is invariant to a systematic offset: a rater uniformly pessimistic by two levels (says 1, 2, 3 where the truth is 3, 4, 5) attains perfect correlation while never guessing the right number. We therefore report rank correlation (Kendall's  $\tau_b$ ) *descriptively only* and base inference on agreement.
- **Transparency of magnitude.** Unlike Spearman,  $\kappa$  is chance-corrected but sensitive to marginal prevalence. We therefore report the full  $6 \times 6$  confusion matrix, the marginals, the mean signed difference (bias direction and size) and the mean absolute error alongside.
- **Within-rater consistency (secondary).** Agreement *among* raters via ordinal Krippendorff's  $\alpha$  (or ICC(2,  $k$ )), separating "wrong together" (shared bias) from "just noisy" (low inter-rater consistency).
- **"Leave-one-rater-out"** A procedure to ensure that our conclusions hold up and that it was not just shifted up and down by certain notorious pessimists or optimists in our rater set.
- **No abstentions.** Every rating occupies a scale position, so nothing is dropped and no missing-value handling is needed.

Related to H2 and H3.

- **Router perspective.** The rating is translated into a binary routing decision with no arbitrary numeric mapping:
  - Human *local* = “Likely”, “Very likely” or “Extremely likely” (levels 3–5, i.e.  $p \geq 0.5$ ).
  - Human *cloud* = “Extremely unlikely”, “Very unlikely” or “Unlikely” (levels 0–2, i.e.  $p < 0.5$ ).
  - Ground truth *local* =  $\hat{p} \geq 1/2$ .
- **Test.** Both sides judge the same prompts, so this is a paired comparison of two proportions  $\rightarrow$  **McNemar test**, one-sided at  $\alpha = 0.025$  with 80 % power ( $\beta = 0.2$ ). H2 is supported if participants assign *local* less often than the ground truth, and we report the magnitude. H3 is supported if participants assign *local* more often than the ground truth, and we report the magnitude.
- **Aggregation** ( $k_h = 3$ ). The per-prompt binary label is the majority vote over the three binarised rater decisions. With an odd number of raters and no abstention category, ties cannot occur; every one of the  $n$  prompts carries a committed decision and none are dropped.

*Exploratory: rater-level quality and its predictors.* The confirmatory analysis aggregates over raters, but the collected demographics let us ask, exploratorily, *which* raters drive (dis)agreement. For each rater we compute an agreement score against  $\hat{p}$  (linearly weighted  $\kappa_w$  and MAE over their 28 prompts) and a signed-bias score (mean signed difference, capturing cloud- vs local-leaning). We then (i) flag outlier raters — unusually accurate or unusually poor — via robust fences (e.g. IQR /  $\pm 2$  robust SD) on the agreement score, and (ii) regress per-rater agreement and signed bias on age, education, occupation, and GenAI usage frequency. This is a correlational, hypothesis-generating analysis on data collected anyway, further justifying a demographic block. With  $N = 30$  raters it is severely underpowered, so we report effect sizes with confidence intervals and make no confirmatory claim: even the a priori plausible “more usage  $\Rightarrow$  better judgment” is reported as a weak association only. This also drives the leave-one-rater-out check — if trimming flagged outliers moves the per-prompt aggregates materially, we report full and trimmed results.

| $\kappa_{\min}$ (floor) | min. detectable $\kappa_w^*$ | inconclusive band | SE at $\kappa_w^*$ |
|-------------------------|------------------------------|-------------------|--------------------|
| 0.10                    | $\approx 0.00$               | (0.00, 0.10)      | 0.040              |
| 0.15                    | 0.05                         | (0.05, 0.15)      | 0.040              |
| 0.20                    | 0.10                         | (0.10, 0.20)      | 0.040              |
| 0.25                    | 0.15                         | (0.15, 0.25)      | 0.041              |
| 0.30                    | 0.20                         | (0.20, 0.30)      | 0.041              |
| 0.35                    | 0.25                         | (0.25, 0.35)      | 0.040              |
| 0.40                    | 0.30                         | (0.30, 0.40)      | 0.038              |
| 0.45                    | 0.36                         | (0.36, 0.45)      | 0.038              |
| 0.50                    | 0.41                         | (0.41, 0.50)      | 0.037              |
| 0.55                    | 0.46                         | (0.46, 0.55)      | 0.035              |
| 0.60                    | 0.52                         | (0.52, 0.60)      | 0.033              |
| 0.65                    | 0.57                         | (0.57, 0.65)      | 0.031              |
| 0.70                    | 0.63                         | (0.63, 0.70)      | 0.028              |
| 0.75                    | 0.69                         | (0.69, 0.75)      | 0.025              |
| 0.80                    | 0.75                         | (0.75, 0.80)      | 0.021              |
| 0.85                    | 0.80                         | (0.80, 0.85)      | 0.018              |
| 0.90                    | 0.86                         | (0.86, 0.90)      | 0.016              |

Table 1. H1 power at  $n = 280$ : minimum true  $\kappa_w$  detectable as below the floor at 80 % power (one-sided  $\alpha = 0.05$ ).

## 2.7 Power analysis

Confirmatory power is carried by the number of prompts ( $n = 280$ ), not the number of raters:  $k_h = 3$  fixes only the precision of the per-prompt aggregate, not the degrees of freedom of the tests. All power figures below are obtained by a seeded simulation released with the preregistration (6000 Monte Carlo replicates; seed fixed).

*H1 (confirmatory – weighted  $\kappa_w$ ).* The threshold test  $H_0: \kappa_w \geq \kappa_{\min}$  vs.  $H_1: \kappa_w < \kappa_{\min}$  (one-sided  $\alpha = 0.05$ ) has no closed-form power, so we estimate it by simulation: (ground-truth, human) prompt pairs are drawn from a latent bivariate normal, binned into the six ordinal levels, with the latent correlation tuned so that the population  $\kappa_w$  hits each target value. At  $n = 280$  the sampling SD of  $\hat{\kappa}_w$  is small (0.016–0.041 across the scale). With a floor of  $\kappa_{\min} = 0.50$  the test has  $\geq 80\%$  power to reject the floor whenever the true  $\kappa_w \leq 0.41$ , and effectively 100% power in the “poor” range ( $\kappa_w \leq 0.20$ ) that H1 anticipates. Only the narrow band  $\kappa_w \in (0.41, 0.50)$  is underpowered – the same inconclusive zone already flagged for H1, where failing to reject is not evidence of adequacy. The simulation treats the human aggregate as a single one of the six levels (median / rounded mean), matching the  $\kappa_w$  computation. The ground-truth marginals are known from the labelling run before any human data are collected, and the simulation is re-run with the observed marginals prior to recruitment; the updated figures are reported with the preregistration.

*H2/H3 (exploratory – McNemar).* The paired binary test (one-sided  $\alpha = 0.025$ , 80% power,  $n = 280$ ) detects a net directional routing imbalance whose size depends on the total discordance rate  $d = p_b + p_c$  between human and ground-truth binary decisions. Solving the asymptotic McNemar sample-size relation for the minimum detectable net difference  $|p_b - p_c|$  gives  $\approx 5.3$  percentage points of all prompts at  $d = 0.10$ ,  $\approx 7.5$  pp at  $d = 0.20$  ( $\sim 21$  prompts net, out of  $\sim 56$  discordant), and  $\approx 9.2$  pp at  $d = 0.30$  ( $\sim 26$  prompts net). A systematic cloud- or local-leaning bias smaller than this would not be reliably detected, which is acceptable given H2/H3 are exploratory and non-gating.

*Rater-level exploratory analysis.* The per-rater regressions of agreement and signed bias on demographics are, with  $N = 30$  raters, severely underpowered by design; we make no power claim and report effect sizes with confidence intervals only.

## 2.8 Topics

Questions are drawn from a topic-stratified test split of MMLU Pro<sup>1</sup> and converted to an open-ended form: the answer options are removed so that the question stands on its own, and the text of the correct option serves as the reference answer. Items originally phrased as “Which of the following...” are retained but tracked as a separate stratum, since they may presuppose a list structure, and natural conversations may indeed be (at least) partly made up of similar queries.

We draw 20 questions per category across the 14 MMLU-Pro categories, yielding  $n = 280$  prompts. Within each category, sampling is a stratified random draw on the  $k = 5$  success rate from the initial labelling run (six bins: 0/5 through 5/5), additionally stratified by the “Which of the following” flag. Allocation across bins is balanced rather than proportional: In the case the pool is strongly polarised or skewed, a proportional draw would leave regions where routing judgements are non-trivial almost unrepresented. Where a (category  $\times$  bin  $\times$  flag) cell cannot be filled, the shortfall is redistributed to the nearest non-empty bin within the same category, and the realised allocation is reported. Prompt order is randomised per participant.

<sup>1</sup><https://huggingface.co/datasets/TIGER-Lab/MMLU-Pro>

*MMLU PRO Benchmark.* We tested our observed Model with the benchmark and received a score of 0.59. In comparison, Claude Opus 4.8 thinking received a score of 0.873<sup>2</sup>

## 2.9 Ground Truth

Ground truth is established in two passes. In the first, every item in the MMLU-Pro test split (12,032 items) is answered  $k = 5$  times by the local model<sup>3</sup> under deployment settings (temperature 1.0, top- $p$  0.95, top- $k$  64, zero-shot). Each response is graded correct or incorrect by a 2-of-3 majority vote of an ensemble of three LLM judges from different model families (the local model’s own family is excluded to avoid self-evaluation bias). This pass fixes the strata used for sampling and supplies the training data for the downstream classifier.

In the second pass, the 280 sampled study prompts — and only those — are re-run for a further 15 generations under identical settings with fresh seeds, and graded by the same judge ensemble. The five original runs are pooled with the fifteen new ones, giving  $\hat{p}$  estimated from  $k = 20$  generations for every prompt in the study. Before pooling we verify that the success rate of the original five runs is consistent with that of the fifteen new ones; if the serving stack has changed, the original runs are discarded and  $\hat{p}$  is estimated from the new generations alone. This second pass costs  $280 \times 15 \times 3 \approx 12,600$  additional judge calls, under 8% of the first pass.

The continuous  $\hat{p} \in [0, 1]$  is mapped to the same six-level scale the raters use, by equal-width sixths: level  $\ell = \min(\lfloor 6\hat{p} \rfloor, 5)$ . The binary label is **locally solvable** if  $\hat{p} \geq 1/2$  (levels 3–5) and **cloud-required** otherwise.

Estimating  $\hat{p}$  from  $k = 20$  rather than  $k = 5$  reduces its standard error from at most 0.224 to at most 0.112, against a scale resolution of  $1/6 \approx 0.167$ . Residual error is therefore mostly confined to adjacent-level slips, which the linear weighting of  $\kappa_w$  penalises at one-fifth weight. Because  $\hat{p}$  nonetheless remains an estimate, we established by simulation the maximum  $\kappa_w$  attainable by a perfectly calibrated rater under this labelling scheme — a rater whose judgement exactly tracks the model’s true latent success probability, scored against a  $k = 20$  realisation of it. That bound is  $\approx 0.87$ . The adequacy floor  $\kappa_{\min} = 0.50$  is set well below it, so a rejection of  $H_0$  cannot be attributed to label noise; observed  $\kappa_w$  is reported both raw and as a fraction of the bound. Items whose true success probability lies near 0.5 remain genuinely borderline and may fall on either side of the binary threshold; we treat disagreement on these as a property of the task rather than as rater error, and report the confusion matrix in full.

To validate the judge ensemble itself, we randomly sample  $z = 100$  prompt judgments (stratified on agreement scores) and re-adjudicate them through the same human setup — three expert raters with access to the original question, the reference answer, and a majority-based final decision — with auditable records of the verification.

The ground-truth prevalence follows directly from the labelling run and need not be estimated in advance. Should the locally solvable class turn out to be too sparsely populated to support an informative test, a preregistered rule switches the split to the median of  $\hat{p}$ , and the fixed threshold  $\hat{p} \geq 1/2$  is reported as a sensitivity analysis instead. This contingency relies solely on model data and requires no prior data collection with participants.

## 2.10 Task

The task is a *prediction* task, not a question-answering task: participants never answer the prompts themselves and never see a model’s answer. For each prompt they make a single instinctive judgement of how likely the named local model is to answer it satisfactorily, on the six-point scale defined under *Response scale*.

<sup>2</sup><https://huggingface.co/spaces/TIGER-Lab/MMLU-Pro>

<sup>3</sup><https://huggingface.co/google/gemma-4-E2B>

- **Prior framing.** By the time the rating task begins, participants have received the neutral local-vs-cloud framing, the definition of a *satisfactory* answer, and any additional meta-level information about the questionnaire. No further conceptual instruction is given mid-task.
- **Definition of “satisfactory” (shown once, up front).** Every prompt has a known correct answer. An answer is *satisfactory* if it states that answer, which is exactly the criterion the LLM-judge ensemble applies: three judges, each given the reference answer, deciding by 2-of-3 majority whether the response matches it. Wording, length, and style do not enter the judgement. Participants are shown the definition and a short illustration that differences in phrasing do not affect correctness, so the threshold is anchored before any rating is made.
- **Stimulus (one prompt per screen).** The open-ended MMLU-Pro prompt is shown on its own: answer options removed, no reference answer, no model output, no category label, no  $\hat{p}$ . Items retained from the “Which of the following...” stratum are shown verbatim.
- **Withheld by design.** To keep the judgement instinctive and to prevent participants inferring the task structure, the interface never reveals the reference answer, the model’s correctness,  $\hat{p}$ , other raters’ judgements, or any running tally or class balance. Withholding the reference answer is intentional: it keeps the judgement about *perceived model capability on the prompt* rather than about a shown answer, consistent with the expert-benchmark difficulty noted in Limitations.
- **Elicited judgement (with every prompt).** The verbatim question is the one given under *Response scale*.
- **Response instrument.** The six-point, midpoint-free scale from *Response scale*, each level carrying its percentage band: *Extremely unlikely* (0–17 %), *Very unlikely* (17–33 %), *Unlikely* (33–50 %) | *Likely* (50–67 %), *Very likely* (67–83 %), *Extremely likely* (83–100 %). The bands are explained in the task instructions before the first rating. Equal band widths are what license the linearly weighted  $\kappa_w$ . The response is forced — there is no “I don’t know” / abstention option and no neutral midpoint — so every prompt receives a committed rating and none are dropped.
- **Trial set and order.** Each participant rates 28 prompts, two per category across the 14 MMLU-Pro categories; prompt order is randomised per participant and navigation is forward only (disclosed beforehand).
- **No feedback.** No correctness feedback is given at any point; learning is not the object of study, only raw judgement.
- **Attention check.** At least one instructed-response item is embedded in the rating stream, and its position is randomised. A failed check is an exclusion criterion (see *Planned Sample*).
- **Session.** The task is completed individually and self-paced in a single session; at two prompts per category the rating block is short (see *Planned Sample*), at an expected median length of 30 minutes.
- **Materials.** Mockups of a sequence of participant-facing screens is presented in Appendix A.

## 2.11 Planned Sample

The number of participants follows from

$$N = \frac{n_{\text{prompts}} \times k_h}{\text{prompts per participant}} = \frac{280 \times 3}{28} = 30,$$

where  $k_h = 3$  is the number of raters per prompt. As noted under *Power analysis*,  $k_h$  does not drive confirmatory power (it only sets the precision of the per-prompt aggregate). At 28 prompts per participant (2 per category), a single

session stays short. In total this yields  $280 \times 3 = 840$  item-level judgements, with every prompt receiving exactly three independent ratings.

Participants are recruited via university and personal networks and screened for prior LLM use. Exclusion criteria are fixed in advance (user or system-side): a failed attention check, an incomplete rating set, and malformed or duplicate responses. Excluded participants' items are re-collected so that three ratings per prompt are preserved.

### 3 Procedure

- (1) **Consent.** Participants are informed about the procedure and what is expected of them.
- (2) **Framing.** Brief, neutral explanation of local vs. cloud models and of the rating task.
- (3) **Rating task.** A stratified sample of open-ended prompts, each rated on the 6-point scale; no feedback given, as learning is not the focus of the user study, but rather raw performance.
- (4) **Post-task.** Attention check, optional free-text reflection.
- (5) **Post-study.** Demographic questions.

### 4 Deferred Analysis Plan: Classifier–Human Comparison

The confirmatory claim of this study, that human raters are unreliable routers, motivates an automated router in their place. Building and evaluating that classifier is the subject of a follow-up study and is *not* analysed here. We nonetheless fix its evaluation plan in advance, because the human side of the eventual comparison is locked by this preregistration while the classifier side is not: architecture, features, hyperparameters, decision threshold, and choice of headline metric all remain free once the human results are known. Registering the following four commitments now keeps “the classifier outperforms human raters on identical material” a testable claim rather than a selected outcome.

- **Single-use rule for the gold set.** The 280 study prompts, labelled at  $k = 20$ , constitute a held-out gold standard. They are excluded from classifier training entirely. All development, model selection, and hyperparameter tuning are performed on a validation split carved from the remaining 11,752 items (stratified by category), whose labels remain at  $k = 5$ . The gold set approximating real world performance with the least noise out of the ground truth data is evaluated exactly once, on the final model, after all design choices are frozen.
- **Primary metric.** Linearly weighted  $\kappa_w$  between the classifier’s predicted level and the  $k = 20$  ground-truth level, on the same six-level equal-width scale used for the human raters, so the two are directly comparable. Standard train/test performance on the category- stratified split is reported separately and is not the basis of the comparison.
- **Comparison test.** Classifier and human  $\kappa_w$  are computed on the same 280 prompts and are therefore paired. The difference  $\kappa_w^{\text{clf}} - \kappa_w^{\text{human}}$  is tested by a bias-corrected accelerated bootstrap over prompts (10,000 resamples), two-sided at  $\alpha = 0.05$ , with the confidence interval reported alongside the point estimate. Resampling prompts captures uncertainty due to which prompts were drawn, but not uncertainty due to which raters were recruited; since confirmatory power here derives from prompts rather than raters, we accept this and note it as a limit on generalisation across rater pools.
- **Threshold rule.** The classifier’s binary routing cut is fixed at predicted  $\hat{p} \geq 1/2$ , matching the ground-truth definition, and is not tuned on the gold set. Any alternative operating point is reported as a sensitivity analysis, clearly labelled as such.

This plan constrains only the evaluation. The classifier’s design is deliberately left open, and nothing in this section should be read as a commitment to a particular modelling approach.

## 5 Limitations

- **Benchmark contamination:** The local model’s training cutoff postdates the MMLU-Pro release;  $\hat{p}$  may therefore overstate the model’s naturalistic capability — the ground-truth labels should be read as an upper bound on local solvability, as is the case for every model, such as Gemma4-e2b , reporting performance on open benchmarks that predate their cutoff date [2].
- **Representativeness:** MMLU-Pro is an academic benchmark and is not representative of naturalistic consumer prompts. It is also an expert benchmark, so the difficulty of understanding the questions might impact user scores on the routing of prompts that do not concern their own day-to-day usage (meaning, if H1 is verified, it is verified for the specific MMLU set, and for prompts the users haven’t written themselves, and therefore likely would not need to route themselves). As such, a replication on real conversational data is left to future work.
- **Generalisability across users:** The claim concerns “people”, but rests on  $N = 30$  participants recruited from university and personal networks and screened for prior LLM use. Confirmatory power derives from the number of prompts, so the design generalises primarily across prompts for a given rater pool; generalisation across the user population is correspondingly limited.
- **Residual label noise:**  $\hat{p}$  is estimated from  $k = 20$  generations and remains an estimate. Items whose true success probability lies near the 0.5 routing threshold may be labelled on either side; a perfectly calibrated rater would still disagree with the label on some of them. The attainable ceiling on  $\kappa_w$  is therefore  $\approx 0.87$  rather than 1, and observed agreement should be read against that bound.

## References

- [1] Aaron Chatterji et al. *How people use chatgpt*. Tech. rep. National Bureau of Economic Research, 2025.
- [2] Gemma Team, Google DeepMind. *Gemma 4 Technical Report*. Tech. rep. Model card: [https://ai.google.dev/gemma/docs/core/model\\_card](https://ai.google.dev/gemma/docs/core/model_card). Google DeepMind, 2026.
- [3] Marah Abdin et al. “Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone”. In: *arXiv preprint arXiv:2404.14219* (2024).
- [4] Claudio Fanconi and Mihaela van der Schaar. “Cascaded language models for cost-effective human-ai decision-making”. In: *Advances in Neural Information Processing Systems* 38 (2026), pp. 11559–11593.
- [5] Yasmin Moslem and John D Kelleher. “Dynamic model routing and cascading for efficient LLM inference: A survey”. In: *arXiv preprint arXiv:2603.04445* (2026).
- [6] J. Richard Landis and Gary G. Koch. “The Measurement of Observer Agreement for Categorical Data”. In: *Biometrics* 33.1 (1977), pp. 159–174. DOI: [10.2307/2529310](https://doi.org/10.2307/2529310). URL: <https://doi.org/10.2307/2529310>.

## A Participant-Facing Materials

This appendix reproduces the instrument as participants encounter it. It is included in the preregistration because several design commitments made in the body — the neutral local-vs-cloud framing, the correctness-only definition of a satisfactory answer, the equal-width bands that license the linear weighting of  $\kappa_w$ , and the information withheld during rating — are properties of the wording rather than of the analysis, and cannot be verified from the analysis plan alone. The screens are shown in the order participants see them. All mockups render the specification in §2.10; the prompts and answers appearing in them are illustrative and are not sampled study items.

## A.1 Consent and Framing

**Before you begin**

You are invited to take part in a study about how people judge what small AI models can do. Participation is voluntary and you may stop at any time without giving a reason.

|                                                |                                     |                                                        |
|------------------------------------------------|-------------------------------------|--------------------------------------------------------|
| What you'll do<br><b>Rate 28 short prompts</b> | How long<br><b>About 15 minutes</b> | Data collected<br><b>Ratings, timing, demographics</b> |
|------------------------------------------------|-------------------------------------|--------------------------------------------------------|

Responses are stored without identifying information and reported only in aggregate. You will not be asked to answer the prompts yourself, and there are no right or wrong responses.

You are invited to participate in the online study which investigates knowledge about local model capabilities. The study is conducted by [REDACTED] and supervised by [REDACTED]. The study with estimated 30 participants will take place in the period from 2026-08-01 to 2026-08-15. Please note:

- Your participation is entirely voluntary and can be discontinued or withdrawn at any time
- For the evaluation, we collect some basic demographic personal information (e.g., age, gender, etc.)
- The study will last ca. 15 minutes
- You have no direct benefit from participating in the study (unless you receive 0.5 VP hours as a student [REDACTED], but you support our work and help to advance research in this area.
- During the course of the session, all responses entered into the system will be meticulously documented, inclusive of timestamp data.
- Recordings and personal data are treated with confidentiality and will be fully anonymized, stored, evaluated, and potentially published so that no conclusions can be drawn about individual persons anymore

The option to decline participation is available. For any inquiries, concerns, or complaints regarding the informed consent process or your rights as a research subject, please contact [REDACTED]. Please read the following information carefully and take the time you need.

☐ I have read the information above and agree to participate.

**Continue**

(a) Consent screen

**Two places an AI model can run**

**On your device**

A smaller model runs locally. Nothing leaves your machine, and it uses less energy.

**In the cloud**

A larger model runs on remote servers. Your request is sent over the internet.

Both kinds of model answer the same sorts of questions. Which one suits a given request depends on the request.

In this study you will see prompts one at a time and estimate how likely a specific local model, **Gemma4-e2b**, is to answer each one satisfactorily. You will not answer the prompts, and you will not see the model's answers.

**Back** **Continue**

(b) Framing screen

Fig. 1. Consent and framing screens. On the consent screen participants are told the task involves rating prompts rather than answering them, and that no right or wrong responses exist; the stated duration is a placeholder pending the pilot. The framing screen describes local and cloud execution in parallel and without evaluative comparison: neither is presented as more capable, and no indication is given of how often either would be appropriate. This neutrality is what keeps the directional hypotheses H2 and H3 uncontaminated by the instructions.

## A.2 Definition of a Satisfactory Answer

**What counts as a satisfactory answer**

Every prompt in this study has a known correct answer. An answer is **satisfactory** if it gives that answer.

Think of it as a panel of experts deciding by majority, each with the correct answer in front of them, asked only one thing: does the model's response say the same thing?

Wording, length, and style are not part of the judgement. Two answers that say the same thing in different words are both satisfactory. A well-written answer that says something else is not.

Correct answer **4.0 L**

- ☒ The volume doubles, so 4.0 L.
- ☒ Four litres.
- ☐ This follows from the gas laws, which relate volume and temperature.

You will not see the correct answers during the task, and you are not expected to work them out yourself.

[Back](#) [Continue](#)

Fig. 2. Definition screen. “Satisfactory” is defined as correctness against a known reference answer, matching the criterion applied by the LLM-judge ensemble. The three short responses illustrate only that differences in phrasing do not affect correctness; they are constructed examples and are not attributed to the local model, so that no information about its overall accuracy is conveyed.

## A.3 Response Scale and Task Rules

**The rating scale**

Your rating estimates how often the local model would answer a given prompt correctly, if it were asked the same prompt repeatedly.

The six options divide that proportion into equal ranges covering every possibility, three below an even chance and three above.

|                                    |                                |                                    |
|------------------------------------|--------------------------------|------------------------------------|
| <b>Extremely unlikely</b><br>0–17% | <b>Very unlikely</b><br>17–33% | <b>Unlikely</b><br>33–50%          |
| 50%                                |                                |                                    |
| <b>Likely</b><br>50–67%            | <b>Very likely</b><br>67–83%   | <b>Extremely likely</b><br>83–100% |

Answer on instinct. Many prompts are specialist questions from fields you may not know, and you are not expected to work them out. Judging the prompt without knowing the answer is the task.

A few things to expect

- ☐ You cannot return to a previous prompt once you've moved on.
- ☐ Every prompt needs a rating — there is no skip or “don't know” option.
- ☐ You will not see the model's answers or be told whether your ratings were accurate.
- ☐ One item asks you to select a specific option instead of giving a rating. It checks that instructions are being read.

[Back](#) [Start rating](#)

Fig. 3. Scale explanation screen. The six equal-width bands are shown in the same three-above-three-below arrangement used during rating, with the 50 % routing boundary marked. The rules disclosed here (forward-only navigation, forced response, absence of feedback, and the presence of an attention check) are all stated before the first rating.